

Polarisation division multiple access (PDMA) separates signals by the use of different polarisations (horizontal and vertical) of the antennae. Thus, two different signals can use the same frequency without causing any interference, one transmitting a vertically polarised signal and the other transmitting a horizontally polarised signal (Fig. 1). Satellite systems use PDMA scheme.

In fibre-optic system, dual polarisation-quadrature phase shift keying (DP-QPSK) is used for multiplexing, which results in very high data rates over a single optical fibre cable. For this purpose, high-speed data is divided into two slower data streams—one using vertical light polarisation and the other using horizontal light polarisation.

Wavelength division multiplexing (WDM) is conceptually the same as the FDM, except that it involves light signals transmitted through fibre-optic channels in order to cater to the high information-carrying capacity of fibre optic cables (Fig. 1). Very narrow bands of light signals from different sources are combined to make a wider band of light.

Evolution of transmission capacity

Multiplexing of several low data rate channels is done in order to utilise the high data rate transmission capacity of transmission media. Such a system, known as the North American digital telephone system/hierarchy, was developed in the early days of telecommunication for sending several telephone signals over a common transmission medium such as copper line. The system uses pulse-code modulation (PCM) in conjunction with TDM.

The basic digital multiplexing standard established in the United States is called the Bell System Level 1 PCM Standard or the Bell T1 Standard. This standard is used for multiplexing 24 separate voice channels together. As a standard voice telephone signal has a bandwidth of 4kHz, it is sampled at twice the 4kHz frequency, that is, at 8kHz (Nyquist rate).

Each sample is then converted to an 8-bit binary number. This occurs 8000

TABLE I
Telecom Transmission Designations

Standard/medium	Designation	Data rate (Mbps)	Number of voice channels
The North American digital telephone system/copper	DS-1	1.544	24
	DS-2	6.312	96
	DS-3	44.736	672
	DS-4	274.176	4032
SONET (synchronous optical network)/optical fibre	OC-1	51.84	672
	OC-3	155.52	2016
	OC-12	622.08	8064
	OC-18	933.12	12,096
	OC-24	1244.16	16,128
	OC-36	1866.24	24,192
	OC-48	2488.32	32,256
	OC-96	4976.64	64,512
Synchronous digital hierarchy (SDH)/optical fibre	STM-1	155.52	2016
	STM-4	622.08	8064
	STM-16	2488.32	32,256
	STM-64	9953.28	129,024

TABLE II
Optical Transmission Windows

Optical window	Operating wavelength
800-900nm	850nm
1250-1350nm	1310nm
1500-1600nm	1550nm

times per second. Thus, if we multiply 8k samples per second $\times$ 8 bits per sample = 64kbps, we get the standard bit rate for a single voice channel in the North American DSI System, which is 64kbps. Each 64kbps voice channel is designated as digital signalling level 0 or DS-0. Each frame in the 24-channel multiplexer comprises 8 bits per channel $\times$ 24 channels + 1 framing bit = 193 bits.

The total data rate when transmitting 24 channels is determined by 193 bits per frame $\times$ 8000 frames per second = 1.544Mbps = T1 or DS-1 designation (Table I). If four T1 lines are multiplexed together, we get 4 $\times$ 24 channels = 96 channels = 6.312Mbps = T2 or DS-2 designation. Multiplexing seven T2 lines together, we get 7 $\times$ 96 = 672 channels = 44.736Mbps = T3 or

DS-3 designation. If six T3 lines are multiplexed together, we get 6 $\times$ 672 = 4032 channels = 274.176Mbps = T4 or DS-4 designation. Such multiplexed signals over copper lines can reach up to 1-2km only due to signal attenuation.

Exploiting the transmission carrying capability of optic fibre, another standard, known as synchronous optical network (SONET), with designations ranging from OC-1 to OC-192 has been developed (Table I). These optical signals can be transmitted over distances up to 100km without the need of repeaters.

In order to meet the demand for higher bit rates, a digital multiplex hierarchy called the plesiochronous digital hierarchy (PDH) evolved in various parts of the world, differing with one another. These PDHs developed and adopted by Europe, North America and Japan used different tributaries for multiplexing and de-multiplexing. Thus, these imposed compatibility and interoperability issues.

In order to overcome issues with PDH systems, a synchronous digital hierarchy (SDH) system has been developed in which each rate is an exact multiple of the lower rate. SDH is a transport hierarchy based on multiples of 155.52Mbps. The basic unit of SDH is synchronous transmission module level-1 (STM-1) (Table I). Transmission rates of up to 40Gbps can be achieved in modern SDH systems.

Even though OC-192 is the maximum data rate specified by SONET, recent developments in SDH technology allow for transmission as high as 40Gbps.

The existing SONET/SDH network architecture is best suited for voice traffic rather than today's high-speed data traffic. To upgrade the system to handle this kind of traffic is very expensive and hence the need for the development of an intelligent all-optical network is aroused.

The solution is to use frequency multiplexing at the optical level or wavelength division multiplexing (WDM/DWDM). Here, the basic idea is to use different optical carriers or colours to transmit dif-